

# Hasanul Mahmud, Ph.D.

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## Professional Summary

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I am an Assistant Professor in the School of Computing at Weber State University, with research expertise in energy-efficient and secure deep learning systems and the analysis of vulnerabilities in large language models. My work focuses on developing robust neural network architectures and trustworthy AI solutions for edge and cloud environments, connecting theoretical foundations with practical applications. With over five years of teaching and mentoring experience, I teach courses in operating systems, network server administration, and data science algorithms. I am committed to advancing the university's teaching and research mission through high-quality scholarship, student mentorship, and inclusive, practice-oriented education that prepares students for careers and leadership in computing and technology.

## Education

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- **PhD in Computer Science**, University of Texas at San Antonio, 2019–2025
- **M.Sc in Computer Science**, University of Texas at San Antonio, 2019–2023
- **B.Sc in Computer Science**, University of Dhaka, 2012–2016

## Work Experience

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**Assistant Professor**, Weber State University July 2026– Present

- Instructing graduate and undergraduate courses in Computer Science and Cybersecurity, including:
  - CS 3100: Operating Systems
  - CS 5580: Data Science Algorithms
  - CYBR 2310: Cybersecurity and Network Management
- Preparing lab materials, programming assignments, and hands-on exercises to enhance student engagement and practical skills.
- Providing academic guidance and support to students through office hours, lab assistance, and project mentoring.
- Collaborating with faculty to align course content with program objectives.

**Visiting Assistant Professor**, Texas A&M International University August 2025–June 2026

- Instructing undergraduate courses in Computer Science and Engineering, including:
  - CSCE 1337: Object-Oriented Programming
  - ENGR 2305: Principles of Electrical Engineering
  - ENGR 2105: Principles of Electrical Engineering Lab

- CSCE 1136: Fundamentals of Programming Lab
- CSCE 3335: Networks and Data Communication
- CSCE 4385: AI Security
- CSCE 3301: Algorithms and Data Structures
- Preparing lab materials, programming assignments, and hands-on exercises to enhance student engagement and practical skills.
- Providing academic guidance and support to students through office hours, lab assistance, and project mentoring.
- Collaborating with faculty to align course content with program objectives.

**Postdoctoral Research Fellow**, University of Texas at San Antonio.    June 2025 – August 2025

- Designing task-oriented conversational agents to assist cancer patients with appointment scheduling, medication adherence, and symptom tracking.
- Evaluating dialogue system performance across NLU, policy management, and response generation for healthcare-specific intents.
- leading a study on assessing chatbot systems in clinical communication scenarios with a focus on safety, interpretability, and patient experience.
- Collaborating with interdisciplinary teams across AI, healthcare, and human-computer interaction for scalable deployment of secure NLP systems.

**Graduate Research Assistant**, University of Texas at San Antonio.    Jun 2020 – May, 2025

- Conducted research on energy-efficient, low-latency deep learning frameworks for deployment on edge devices such as Raspberry Pi and Jetson Nano.
- Designed robust DNN architectures and defense strategies against adversarial attacks (e.g., FGSM, PGD, backdoor), contributing to multiple first-author publications in top-tier conferences (IPDPS, IEEE Cloud, ICPR).
- Proposed and implemented converting autoencoder-based architectures to improve DNN classification accuracy under SLO and energy constraints.
- Led benchmarking and profiling of inference latency, energy usage, and throughput of deep learning models across embedded and cloud platforms.
- Mentored undergraduate and MS students in research methodology, experimental setup, and model evaluation; provided feedback on their theses and paper drafts.
- Supported lab operations by configuring hardware/software stacks, documenting reproducible workflows, and assisting in preparing grant proposals (e.g., NSF submissions).

**Graduate Teaching Assistant**, University of Texas at San Antonio.    Aug 2019 – May, 2025

- Assisted and independently led lectures, labs, and assignments for undergraduate and graduate computer science courses.
- Created and graded assignments, projects, and exams; managed in-person and online students.
- Contributed to syllabus and curriculum design for core systems and AI-related courses.

- Mentored students in algorithm design, programming, and high-performance computing frameworks.

- **Courses Assisted:**

- Spring 2025 – High-Performance Machine Learning, Analysis of Algorithms
- Fall 2024 – Parallel Computing
- Spring 2024 – Parallel Algorithms
- Spring 2023 – High-Performance Computing
- Fall 2022 – Parallel Computing
- Spring 2022 – High-Performance Computing
- Summer 2022 – Cloud Computing
- Summer 2021 – Application Programming
- Spring 2021 – High-Performance Computing
- Summer 2020, Spring 2020 – Discrete Mathematics
- Fall 2019 – Computer Programming

**Software Engineer**, Placovu Inc., Dhaka, Bangladesh

2017 – 2018

- Built backend systems using Java and C#.
- Optimized search performance using Elasticsearch and SQL Server.

## Technical Skills

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**Languages:** Python, Java, C++, R, SQL, Rust

**Frameworks & Tools:** TensorFlow, PyTorch, Keras, Docker, Git, OpenMP, MPI, CUDA, PySpark, Hadoop

**Visualization:** Tableau, Power BI, Seaborn, Matplotlib

**Hardware/Testbeds:** Raspberry Pi 4, Jetson Nano, Chameleon Cloud, DGX Cluster, Stampede

## Grant Writing Experience

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- **Defense Advanced Research Projects Agency (DARPA), Ongoing:** Actively contributing to the development of a research proposal addressing security vulnerabilities in Large Language Models (LLMs) within cloud infrastructure environments. The proposed work focuses on enhancing cloud resilience by designing methods to detect, analyze, and mitigate LLM-related security threats, including adversarial attacks and model exploitation.
- **National Science Foundation (NSF), 2021:** Submitted a research proposal focused on geolocation using Earth imagery. The project aimed to identify geographic locations by analyzing visual features such as skylines and ridgelines. Developed the technical methodology for location detection using image-based feature extraction and pattern matching techniques.

## Selected Publications

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- \*A Converting Autoencoder Toward Low-latency and Energy-efficient DNN Inference at the Edge\*, IPDPS 2024 Paper Link

- **\*CAE-Net: Enhanced Converting Autoencoder based Framework for Low-latency Energy-efficient DNN with SLO-constraints\***, IEEE Cloud Summit 2024 [Paper Link](#)
- **\*EncodeNet: A Framework for Boosting DNN Accuracy with Entropy-driven Generalized Converting Autoencoder\***, ICPR 2024 [Paper Link](#)

## Under Preparation/Submission

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- **Efficient Detection of Backdoor Attacks** – Designed a framework to detect backdoor attacks by perturbing inputs and analyzing model activations and entropy.
- **Energy Consumption Benchmarking of Individual Layers in Deep Neural Networks** – Conducted detailed experiments to measure energy usage, memory consumption, CPU utilization, and latency of each layer in various DNNs on Raspberry Pi and Jetson Nano.
- **Vulnerability Analysis in Deep Neural Networks** – Evaluated widely-used DNN models under both white-box and black-box attacks. Analyzed activation patterns, entropy, and saliency maps using Explainable AI techniques to identify vulnerabilities across layers.
- **A Survey on Vulnerabilities of Large Language Models in Healthcare Data Applications** – Surveys vulnerabilities of large language models in healthcare data applications, identifies gaps in recent advancements, and proposes new research directions to strengthen security and reliability.

## Academic Service

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- Technical Reviewer, Supercomputing Conference (SC), 2025
- Technical Reviewer, SaT-CPS, 2021
- ABET Course Design and Evaluation, Fall 2025, Spring 2026
- Technical Committee Member, ICMLA 2026

## Research Projects

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- **Modular Autoencoders for Improving DNN Classification Accuracy** – Explored clustering methods to capture intra- and inter-class relationships for hierarchical DNN design. Proposed modular architectures to reduce the memory footprint.
- **Designing Memory-Aware Frameworks for Tiny Edge Devices** – Developed multiple lightweight DNN models with fewer than 20K parameters and <84KB size, achieving over 80% accuracy on CIFAR-10, enabling deployment on highly constrained devices.
- **Adversarial Vulnerability Analysis of Deep Neural Networks** – Studied attacks (PGD, FGSM, CW, Trojan) on VGG, ResNet, ShuffleNet. Tools: PyTorch, TensorFlow, Jetson Nano, Raspberry Pi.
- **Energy-Aware DNN Benchmarking on Edge Devices** – Layer-wise energy measurement; proposed energy-aware pruning.

## Academic Projects

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- **Distributed Healthcare System via Java RMI** – Built a secure backend system for distributed healthcare data.
- **Operating System Caching Mechanism** – Simulated caching to evaluate page hits/misses.

- **Animated Road Traffic Simulation** – Real-time simulation of cars/pedestrians.
- **E-Commerce Shopping Platform** – Online shop with full cart/order/user interface.
- **Fruit Ninja Game** – Built an OpenGL/OpenCV fruit slicing game.
- **Library Management System** – Book tracking system with user/librarian interface.
- **Line-Following Robot with Bluetooth Control** – Robot navigation and manual override via Bluetooth.
- **Police Department Database System** – Centralized personnel tracking system with web UI.

## Conferences & Workshops

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- IEEE Cloud Summit 2024 (Presenter)
- IPDPS PhD Forum 2022 (Presenter)
- UTSA Graduate Research Symposium 2022 (Presenter)
- MVAPICH User Group Conference, 2022 at Ohio State University (Presenter)
- NSF AI Spring School, 2024 at UTSA (Presenter)

## Honors & Awards

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- NSF Travel Grant — MUG 2022
- TCPP Award to Attend – IPDPS 2024
- Finalist — Radiance Innovation Bowl 2023
- UTSA Graduate School Professional Development Award 2024

## Certifications

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- HackerRank: Problem Solving (Intermediate) [Link](#)
- Solved 350+ Algorithmic Problems on LeetCode. [Link](#)